Name: ____________

Date: ____________

Linear Modeling & Residuals

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can calculate and interpret average rate of change with appropriate units.**

RELATED SKILLS

- D — Calculate average rate of change over an interval.
- D — Interpret average rate of change with contextual units.
- A — Attach and interpret units for rates, inputs, and outputs.

☐ **2. I can construct a linear function from two data points, a table, a graph, or a context.**

RELATED SKILLS

- A — Calculate slope from two points.
- A — Explain a model parameter in the language of its context.
- R — Convert among common forms of a linear equation.

☐ **3. I can fit a linear regression model to data using technology and interpret its parameters.**

RELATED SKILLS

- I — Recall and correctly use regression terms including explanatory variable, response variable, correlation, residual, and residual plot.
- I — Use technology to determine a least-squares linear regression model.
- D — Explain a model parameter in the language of its context.

☐ **4. I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.**

RELATED SKILLS

- I — Interpret the sign and magnitude of a correlation coefficient in context.

☐ **5. I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.**

RELATED SKILLS

- I — Calculate an observed value minus its model-predicted value.
- I — Construct and interpret a residual plot.
- I — Use residual patterns and contextual evidence to judge whether a model is appropriate.

☐ **6. I can use a linear model to make and interpret predictions, including solving for either variable.**

RELATED SKILLS

- D — Solve for an input associated with a specified output.
- A — Identify input and output values in a formula, graph, table, or context.
- A — Restrict a model to meaningful contextual inputs.